HoloMassage XR: An XR Learning System for Beginning Massage Training

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I. Introduction

1.1 Observation

1.1.1 Learning Process Observation Based on a Massage Therapist Interview

We conducted a semi-structured interview of approximately 20 minutes with a female massage therapist currently working at Magic Herb. During the interview, we asked questions to guide her in recalling and describing her learning experiences during her apprenticeship.

(1) Identification of Acupoints and Meridians

She mentioned that her learning experience mainly centered on mentor demonstration and explanations. The mentor would usually point out a specific acupoint, explain its function, and demonstrate basic pressing techniques. In class, training mannequin and printed body diagrams were often used to help apprentices recognize major meridians and common acupoints. However, these approaches often keep apprentices focused on isolated acupoints and local functions, rather than helping them understand the body as an interconnected whole.

During training or assessment, when the mentor assigned an area that needs treatment, apprentice often feel uncertain. She knew some general acupoints and meridians, but did not know how to infer the relevant massage pathway. For example, some conditions may require pressing distal acupoints, while others may involve reflex zones on the hands and feet. However, these connections cannot be mastered simply by understanding a few individual points.

In this regard, current teaching methods are unable to effectively help her fully identify the complete network of meridians and acupoints, nor can they help her organize her fragmented knowledge of meridians and acupoints into a holistic understanding of the body. Ultimately, she found it difficult to start from the problem raised by the mentor and determine the truly relevant massage pathway.

(2) Mentor-Guided Bodily Calibration

The massage therapist also mentioned that during the apprenticeship stage, the mentor usually taught apprentices how to locate acupoints and meridians through embodied exposure. For example, he would directly manipulate a specific meridian on an apprentice's body or press a particular acupoint, allowing them to feel its approximate location. However, in this process, apprentices were more like observers or recipients of the experience, rather than direct participants.

However, this learning process can easily lead apprentices to assume that body structures are standardized. In real assessments, multiple mentors may serve as different subjects and ask apprentices to locate a specific meridian or acupoint on their bodies quickly. Because people differ in muscle condition, body shape, and physical structure, the location of the same meridian may vary from one person to another.

Therefore, in this assessment context, the differences between real bodies and standardized acupoint models often left apprentices uncertain, leading to inaccurate localization and a lack of confidence in their practice. This suggests that acupoints and meridians are not fixed coordinates but relational locations that must be continuously calibrated through interaction with real bodies.

1.1.2 Observation based on Video

Through our interactions with massage therapists, we observed that beginner therapists may face many challenges during early training. To better understand these issues, we further analyzed online educational massage videos.The most helpful resources come from the YouTube channel Bodyology Massage Courses. From this channel, we selected "Deep Tissue Massage Techniques - Massage Courses London" for a detailed analysis.

After observing this video, we noticed that the massage instructor speaks rather quickly. At the same time, his step-by-step breakdowns lack sufficient detail. Moreover, the instructor needs to use skeletal models to aid in explaining acupoints and muscle locations during instruction. However, such knowledge cannot be shown directly on living humans. For example, at about 4:45, the instructor uses a skeletal model to show the erector spinae muscle. As a result, beginners may struggle to fully grasp massage knowledge. At the same time, they need to transfer model-based knowledge to a real person, which can make learning more difficult.

Beyond that, this video highlights two key aspects of massage instruction. First, beginners need to practice with real people to learn massage locations and techniques. Second, live practice is important because students must eventually apply what they learn to real clients.

1.2 Problem of Practice

Based on interviews and video observations, we have identified the following two main practical issues.

1.2.1 Difficulty in linking symptoms to relevant massage pathways

In early training, massage apprentices need to memorize extensive knowledge about meridians and body systems. However, many beginners can only memorize a few key acupoints and major meridians before practice. As a result, they may focus on isolated acupoints without considering interconnection within the human body.

For example, when a client complains about back pain, a beginner may only massage the painful area. However, the pain may also relate to the legs or other meridian points. Because apprentices may not understand these distant associations, they can struggle to identify the right meridian. If the client's symptoms do not improve, apprentices may become more confused. They may not know whether to adjust the pressure or massage other areas. This uncertainty can

also create stress, as beginners may worry that one mistake could make the client ask for another therapist. As a result, it becomes harder for them to apply meridian and acupoint knowledge to real-world decisions.

Therefore, the first issue we identified is that beginner massage therapists struggle to memorize the necessary knowledge. More importantly, they find it difficult to connect that knowledge to clients' symptoms and identify relevant massage techniques. Consequently, we believe XR can serve as a supportive tool for beginners. XR can break down and visualize invisible meridians, acupoints, and anatomical relationships to learners. In this way, XR can support their learning, decision-making, and application of knowledge to real clients.

1.2.2 Positional Miscalibration Across Different Bodies

The human body is dynamic rather than standardized because the placements of meridians, acupoints, and bodily structures vary across individuals. However, in current training practices, apprentices typically learn localization in two relatively fixed ways. First, mentors demonstrate directly on apprentices' bodies, allowing them to feel how an experienced practitioner can quickly and accurately locate acupoints. Second, apprentices use a mannequin, or occasionally treat peers as silent practice subjects, to practice localization and calibration.

From the description above, current teaching methods fail to fully support the development of dynamic localization skills and bodily perception needed in real-life situations. During evaluations, apprentices are often asked to quickly locate a specific body area on different mentors' bodies. In these situations, they often struggle with calibration and lack confidence in their technique. In other words, an apprentice may accurately locate acupoints on a mannequin or a paper body diagram, but when working with different real human bodies, they may not always be able to press the right spot.

II. Thesis

Based on these issues, this paper proposes the HoloMassage XR Learning System. It uses AR visualization, an AI Tutor, and real peer practice to help learners practice basic massage techniques, identify massage pathways based on symptoms, adapt to different bodies, and recognize the body as an interconnected system. As a result, learners can apply meridian knowledge and massage techniques in real-world practice.

III. Design

3.1 Overview

3.1.1 Overview System

The HoloMassage XR Learning System integrates AR glasses, an AR knowledge projection system, an AI tutor, and real peer practice sessions. Through this design, the HoloMassage XR Learning System helps learners address two key challenges. First, it helps them identify relevant meridians, acupoints, and massage pathways based on client symptoms. Second, it supports more precise positioning and adjustment on different bodies.The system has two phases. First, Phase 1 involves learning the fundamentals of meridians and massage.Then, Phase 2 focuses on task-based practice in real contexts. The learning process occurs in a specific order, starting from Phase 1, followed by Phase 2 Task 1 and Phase 2 Task 2.

This design was inspired by Hall and Jurow's (2015) work on conceptual practices. They argue that concepts are not merely "mental structures held by an individual," but are closely tied to the practical activities in which people engage (Hall & Jurow, 2015). Therefore, meridian and acupoint knowledge is not only memorized. Learners first understand it through AR visualization. Then, they apply it through hands-on practice and interactive feedback. Finally, learners gradually turn abstract knowledge into practical judgment in real massage scenarios.

3.1.2 Technology Reflection: Why AR and AI are Used in Massage Training

This design uses AR glasses and an AI tutor, not for the sake of technological novelty, but because the difficulty of learning massage stems largely from the way knowledge is presented. Traditional teaching relies on standard anatomical diagrams, fixed acupoint models, and mentor demonstrations. However, learners have trouble not only identifying the vast number of meridians and acupoints, but also comprehending the body as a linked system. Furthermore, they often encounter miscalibration during practical assessments on real patients due to individual physical variations. Wilensky and Papert (2010) point out that learning difficulties are sometimes not due to a lack of ability on the part of the learner, but rather to an inappropriate way of representing knowledge. Therefore, this design attempts to reconstruct the representational infrastructure of massage learning.

Drawing on Wilensky and Papert's (2010) framework, current representational reconstruction offers key educational advantages. In terms of power, AR overlays can project the otherwise abstract and invisible meridian lines and acupoint grids directly onto a mannequin or a real body, enabling learners to gradually construct understanding through active exploration, comparison, and interaction.

From a cognitive perspective, AR technology adheres to the Spatial Contiguity Principle. It overlays meridian pathways, distal acupoints, and reflex zones directly onto the body, reducing learners' cognitive burden of switching between reference materials and the model. In that case, learners can observe, manipulate, and calibrate simultaneously, making it easier for them to understand the spatial relationships of acupoints and develop accurate localization skills.

On the affective level, AI Tutor provides personalized prompts and on-demand instruction. Compared to one-time demonstrations by a mentor, this interactive and repeatable guidance

reduces anxiety and lowers the entry barrier during the initial learning phase, while enhancing learners' engagement and motivation to explore, thereby encouraging them to take a more proactive approach to understanding massage techniques.

From a social perspective, AR systems transform abstract, difficult-to-describe massage techniques into visual information that mentors, apprentices, and peers can use as a common reference. At the same time, peer practice can be transformed into more goal-oriented collaborative learning activities through the use of highlighted paths, AI tutors, and feedback prompts. In this sense, this representation is easier to step into real-world training scenarios and fosters more effective guidance and collaboration within the learning community.

In terms of diversity, this new approach offers more accessible entry paths for learners with different backgrounds. Rather than presenting all information at once, the system breaks down complex content into smaller units. Learners can first familiarize themselves with the meridian lines and acupoint grids, then move on to localization and calibration exercises on a live human body, and finally develop a comprehensive understanding of the body as a holistic system through more integrated tasks.

Overall, HoloMassage XR does not aim to replace human mentors or physical interaction. By restructuring the representational infrastructure of massage training, it offers a more effective, interactive, and low-pressure learning path. Through AR visualization, AI tutor guidance, practice on mannequins during pre-training, and advanced practice with human partners, learners can develop their ability to identify, locate, and assess the human body. At the same time, this shared visual representation facilitates communication and interaction among the community of practice and helps learners with varying backgrounds gradually transition into practical training.

3.2Target Users:

The HoloMassage XR Learning System is designed as a pre-training tool for adult beginners. Specifically, it supports adult beginners in learning basic massage knowledge and introductory techniques. Importantly, it is not intended for professional certification, clinical massage, or medical training.

3.3 Detailed Design

3.3.1Phase 1:*AR-Guided Meridian Mapping Pre-training*

Phase 1 of the HoloMassage XR Learning System helps learners build a foundational understanding of meridians, acupoints and basic massage techniques. It combines AR glasses, an AR knowledge projection system, and guidance from an AI Tutor. It helps learners move beyond memorizing isolated acupoints and understand the body's meridian system before practicing with real peers. This design is inspired by Keifert et al. (2020), who argue that "the visualization served as a bridge between embodiment and conceptual ideas." In this design, AR projection does not simply display meridian knowledge. Instead, it connects abstract meridians and acupoints to specific body locations.

Before using the system, learners first need to set up a mannequin. When learners wear the AR glasses, the system will project human meridians and acupoints onto it. Meridians will appear as yellow lines, while acupoints will be displayed as red spots. Learners can tap different meridians and acupoints to view their names, specific locations, functions, and related massage techniques for pain relief. The AI Tutor will guide learners through this process.

After each set of meridians and acupoints, the AI Tutor will give a short quiz to check learners' progress. For example, after learners study the Pericardium Meridian, the system will temporarily hide the projection and provide related questions. The AI Tutor will first ask, "Please identify the location of the Pericardium Meridian on the mannequin."Then, the AI Tutor will ask,

“Please point out the key acupoints along this meridian and explain which massage techniques can relieve pain in this area.” If learners answer correctly, the AI Tutor will affirm their answer and briefly review the key knowledge to strengthen learners’ memory. However, if they answer incorrectly, the system will highlight the correct meridian, acupoints and correct massage techniques. Then, the AI Tutor will explain reasons behind and encourage learners to try again.

Through Phase 1, learners can repeatedly study meridian knowledge in a low-pressure setting and build a more complete understanding of the body system.

3.3.2 Phase 2: Task 1 Real-Body Calibration Practice

Task 1 primarily addresses Problem 2: Positional Miscalibration Across Different Bodies. Its core objective is to help apprentices apply the meridian knowledge acquired during pre-training to real human bodies and develop dynamic localization and calibration capabilities.

Distributed Support for Real-Body Calibration

In Real-Body Calibration Practice, the AI Tutor first gives a specific task, such as: “Follow the Pericardium Meridian on your partner’s forearm and find areas that feel tight or nodular.” Then, the AR scaffold highlights the approximate area of the meridian with an oval marker. Within this highlighted range, the apprentice moves, presses, and manipulates the area while adjusting angle and pressure. Based on the partner’s button feedback, the apprentice judges whether the current location is “off target,” “close,” or “on an effective point.”

This design is inspired by Hutchins’ (1995) concept of distributed cognition. In his cockpit example, pilots process speed information through speed bugs, instrument displays, verbal confirmation, and collaboration with others. In this sense, flying is not supported by one pilot’s internal cognition, but by a larger system of people, tools, and the environment.

Building on this idea, our design creates multiple learning mediators that work together to support apprentices' real-body localization and calibration skills. The AI Tutor provides the task, the AR area cue offers external visual guidance, the real body delivers tactile variation, and the Sensation Feedback Button helps apprentices judge whether a location is effective. Together, these elements form a distributed learning system where apprentices coordinate visual cues, bodily sensation, and peer feedback. Through repeated observation, judgment, adjustment, and negotiation in real-body practice, they gradually develop calibration skills.

Reciprocal Learning Through Feedback and Role Exchange

Ma and Munter's (2014) study inspired us to position the real-body partner as an active feedback provider rather than a passive practice subject. In their study of a skateboard park, Karl and Hal practiced the same move together. They took turns trying, observed each other's body movements, and exchanged feedback. For example, Hal noticed that Karl opened his shoulders backward when exiting the ramp, which helped him rotate more effectively. This also helped Karl recognize which movements worked and what he still needed to adjust. This example shows that peer learning can happen through mutual observation, embodied communication, and immediate feedback.

Building on this, our design positions the real-body partner as an active feedback partner. As the apprentice moves within the approximate AR-guided area, the partner gives immediate button feedback, such as "no sensation," "soreness," "too painful," or "sense of release." These responses do more than evaluate comfort; they help apprentices adjust the location, angle, pressure, speed, and direction more effectively.

In this way, peer massage becomes a process of seeking teaching and reciprocal learning. One person practices, the other provides feedback, and then they switch roles. Through repeated

cycles of goal setting, monitoring, feedback, and adjustment, both partners improve their skills through interaction. Ultimately, repeated practice with diverse bodies helps apprentices quickly and accurately locate acupoints, even when body posture, soft-tissue thickness, and muscle tension vary across individuals.

Identity Resources for Becoming a Massage Practitioner

Based on our interview and training video observations, apprentices in current training often remain on a peripheral trajectory. They mainly watch the mentor's demonstrations, practice alone afterward, occasionally use peers as temporary practice subjects, and then wait for periodic assessments. In this structure, apprentices may see themselves as students afraid of making mistakes and waiting to be evaluated, rather than as active participants in a community of practice.

Nasir and Cooks (2009) note that when a learning environment provides sufficient material, relational, and ideational resources, learners are more likely to develop a sense of belonging, cultivate identity, and have greater opportunities for participation, thereby gradually moving toward an inbound trajectory. Therefore, our design provides richer identity resources to support apprentices in transitioning from peripheral participation to more active practical engagement.

First, AR glasses and meridian visualization serve as material resources. They provide apprentices with more balanced and repeatable practice opportunities, helping them develop meridian recognition, location calibration, and a holistic understanding of the body. Second, peer partners and the Sensation Feedback Button serve as relational resources. Through immediate and specific feedback, apprentices can practice, communicate, and improve together. As a result, this supportive relationship can strengthen apprentices' sense of belonging and motivation, making them more willing to keep trying and improving even after receiving negative feedback

from mentors during assessments. Third, the AI Tutor's encouraging task prompts serve as ideational resources. For example, during calibration practice, the AI Tutor might say: "Take your time. Like a real massage practitioner, start from the inner forearm and slowly follow the meridian downward, as if a submarine were sensing the flow of water." This positive guidance helps apprentices reinterpret their role: they are not merely trainees waiting to be assessed, but future massage practitioners developing judgment, tactile sensitivity, and professional confidence.

Overall, our design aims to reorganize the way apprentices acquire identity resources, thereby shifting their participation position and helping them gradually develop practical skills, a sense of belonging, operational confidence, and a professional identity.

3.3.3 Phase 2 : Task 2 Symptom-to-Massage Pathway Training

After learners complete Task 1, the AI Tutor will introduce Task 2. This task helps learners practice massage, locate massage positions, and identify appropriate massage approaches based on specific client symptoms. In this way, it addresses the first problem discussed earlier.

Task 2 is inspired by Lave (1996). Lave argues that learning is not simply about acquiring knowledge. Instead, it is "changing participation in changing communities of practice" (Lave, 1996). Therefore, Task 2 uses scenario-based exercises with live peers to help learners participate in real-world massage judgment and practice.

In Task 2, the AI Tutor will introduce a real peer as the practice subject and present a scenario-based case. The peer may not have actual symptoms. Instead, the AI Tutor will generate a simulated case. For example, the AI Tutor might say, "Your peer is currently complaining of lower back pain. Based on what you've learned and using the AR glasses, identify the relevant meridians and acupoints for massage." Then, learners will see prompts on the peer's body

through the AR glasses. The system will first highlight the painful lower back area. Then, it will show related meridian pathways and distal acupoints, such as the Wei Zhong acupoint (BL40) near the popliteal fossa. This helps learners understand that relieving lower back pain may involve both the lower back and other distal areas.

During the practice, learners will massage the highlighted areas with guidance from the AI Tutor. If they focus only on the painful area or apply insufficient pressure, the AI Tutor will correct them and provide voice prompts. For example, when learners press near the Wei Zhong acupoint, the AI Tutor may say, "This is a common distal trigger point associated with lower back pain. Please confirm the location and gradually increase your pressure."

Through Task 2, learners move beyond massaging only the painful area. Instead, they learn to infer massage pathways from symptoms and apply Phase 1 knowledge in real-person practice.

IV. Risks:

4.1. Artificial Intelligence Risk:

4.1.1 User Privacy and Data Security Issues

The first risk concerns learner privacy and data security. The AI Tutor needs to collect data such as interaction locations and exercise performance to provide personalized guidance to learners. However, learners may worry that their activities are over-monitored or misused. Therefore, HoloMassage XR should clearly explain how data is collected and used, while minimizing unnecessary data collection.

4.1.2 Learning Risks Associated with AI Misjudgments

The second risk is AI misjudgment. The AI Tutor evaluates acupoint accuracy, pressure, and posture. However, beginners may not know whether its feedback is reliable. If the AI gives incorrect guidance, learners may treat mistakes as correct knowledge. At the same time, incorrect

guidance during real peer practice may cause discomfort to the peer. Therefore, the system should clearly state that the AI Tutor only provides basic, low-risk support and cannot replace a professional trainer.

4.2 Equity and Inclusion Risks

This section examines whether the default design of HoloMassage XR may unintentionally favor learners with certain physical conditions, technical experience, and resource support, or increase the environmental pressures faced by some groups.

First, XR devices may cause visual fatigue, dizziness, or physical discomfort. Lee and Hu-Au (2021) note that immersive technologies can raise bodily safety concerns, such as eye strain and motion sickness. Therefore, long-term use of AR glasses may be less accessible for learners with severe myopia or neurological sensitivity, which could limit their sustained participation in the learning community. To reduce this risk, the system should use a gradual training design and allow learners to switch to a non-AR projection mode at any time.

Second, AR technology may also raise the learning threshold. Massage training already requires learners to process meridian locations, bodily sensations, and practical judgment at the same time. If learners must also spend significant effort understanding visual cues, AI instructions, and button controls, the technology itself may create additional cognitive load (Li & Lee, 2024). As a result, learners with less technical experience may focus more on “how to use the system” than on massage practice. To avoid this, the design should keep the interface simple and provide clear onboarding guidance.

In addition, device cost may create unequal access to resources. AI-powered AR glasses with advanced audio and video functions are often expensive, making them more accessible to well-funded training institutions. By contrast, smaller massage schools or learners in areas with

insufficient infrastructure may have fewer opportunities to use the same technology. As a result, a tool designed to expand learning opportunities could instead deepen inequalities (Van Dijk, 2017). To reduce this risk, future designs should offer lower-cost options. For areas with limited electricity or internet access, the system should also support offline use, low-power mode, and local data storage.

Finally, as Yañez-Barnuevo (2025) reports, systems that rely heavily on high-performance data centers may increase energy and water consumption, placing additional pressure on regions with limited environmental resources. Therefore, cloud-based data processing in AR systems could further burden marginalized communities and deepen existing inequalities. To address this risk, future designs should reduce dependence on cloud storage and prioritize more sustainable technical solutions.

V. Conclusion

This paper contends that the value of educational technology lies not in replacing mentors or real-body interaction, but in reorganizing the representational infrastructure, bodily experiences, and social relationships within the learning process. In response to two core issues in existing massage training, namely apprentices' difficulty in deriving appropriate massage pathways from physical symptoms and their difficulty in transferring acupoint localization from standard models to different real bodies, HoloMassage XR serves as a technological solution that supports distributed learning, embodied learning, reciprocal peer learning, and identity development. Through AR visualization, AI Tutor guidance, mannequin-based practice, and live-person interaction, the system expands apprentices' learning and practice opportunities, making the traditionally passive and isolated apprenticeship process more interactive and supportive. The learning process begins with foundational exercises on mannequins and gradually transitions to

live-person calibration exercises and symptom-oriented integrated tasks. In this way, HoloMassage XR helps apprentices transform meridian knowledge into dynamic calibration skills and professional judgment applicable to real-world practice.

Work Cited

Bodyology Massage Courses. “Deep Tissue Massage Techniques - Massage Courses London.” *YouTube*, 8 May 2012, www.youtube.com/watch?v=ON7DivsisIs. Accessed 30 Apr. 2026.

Hall, R., & Jurow, A. S. (2015). Changing concepts in activity: Descriptive and design studies of consequential learning in conceptual practices. Educational Psychologist, 50(3), 173-189.

Hutchins, E. (1995). How a cockpit remembers its speeds. *Cognitive Science, 19*(3), 265–288.

Keifert, D., Lee, C., Enyedy, N., Dahn, M., Lindberg, L., & Danish, J. (2020). Tracing bodies through liminal blends in a mixed reality learning environment. International Journal of Science Education, 42(18), 3093-3115.

Lave, J. (1996). Teaching, as learning, in practice. *Mind Culture and Activity*, *3*(3), 149–164. https://doi.org/10.1207/s15327884mca0303_2

Lee, J. J., & Hu-Au, E. (2021). E3XR: An Analytical Framework for Ethical, Educational and Eudaimonic XR Design. *Frontiers in Virtual Reality*, *2*, 1–14. https://doi.org/10.3389/frvir.2021.697667

Li, X., & Lee, J. J. (2024). Failure stories and surprising findings: Learning from when VR did not work. In M. L. Bourguet, J. M. Krüger, D. Pedrosa, D. Dengel, A. Peña-Rios, & J. Richter (Eds.), *Immersive Learning Research Network: iLRN 2023* (*Communications in Computer and Information Science*, Vol. 1904). Springer. https://doi.org/10.1007/978-3-031-47328-9_2

Ma, J. Y., & Munter, C. (2014). The spatial production of learning opportunities in skateboard parks. *Mind, Culture, and Activity, 21*(3), 238–258. https://doi.org/10.1080/10749039.2014.908219

Nasir, N. I. S., & Cooks, J. (2009). Becoming a hurdler: How learning settings afford identities. *Anthropology & Education Quarterly, 40*(1), 41–61. https://doi.org/10.1111/j.1548-1492.2009.01027.x

Van Dijk, J. A. G. M. (2017). Digital divide: Impact of access. In P. Rössler, C. A. Hoffner, & L. Zoonen (Eds.), *The international encyclopedia of media effects*. Wiley. https://doi.org/10.1002/9781118783764.wbieme0043

Wilensky, U., & Papert, S. (2010). *Restructurations: Reformulating knowledge disciplines through new representational forms*. Constructionism 2010, 1–15.

Yañez-Barnuevo, M. (2025, June 25). *Data Centers and Water Consumption*. EESI. https://www.eesi.org/articles/view/data-centers-and-water-consumption